Etienne **LASALLE**
Ph.D. in Mathematics

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My research focuses on **statistics related to graph-structured data**. During my thesis, I developed and studied tools for *multi-scale graph comparisons* as well as *statistical comparisons of graphs samples*, with applications in neural networks learning.
More recently, my work lies in the field of **signal processing**. I have study various aspects of *graph data compression*. I also exploit graph structures using signal processing for applications in *epidemiology and neuroscience*.

Graphs | Statistics | Signal processing | Computational statistics | Compression | Topological analysis

Positions and research experience.

Mar 25 - Sep 26 **Postdoctoral Position**, SIMS Team, LS2N, Nantes (LS2N funding – CNRS-DIALOG and LS2N funding).
Graph signal processing and graph analysis for epidemiology and neuroscience.
Supervision : Barbara Pascal and Mira Rizkallah.

Jan 23 - Feb 2025 **Postdoctoral Position**, ENS de Lyon (LIP) and Inria (Ockham).
On compressive learning and graph-structured data.
Supervision : Rémi Gribonval and Paulo Gonçalves.

Sep 19 - Dec 22 **PhD Thesis**, Université Paris-Saclay and Inria Saclay.
Some contributions to the statistical analysis of graph-structured data.
Advisors : Frédéric Chazal and Pascal Massart.

Oct 18 - Jul 19 **Pre-Doctoral Research Year**, EPFL, Lausanne, Switzerland.
Probabilistic and statistical studies of topological tools for graph analysis, in the context of Neuro-Topology.
Supervision : Kathryn Hess

2018 **M2 Internship** (4 months), INRIA-Saclay, Palaiseau, France.
Development of anomaly detection methods based on objects and tools from Topological Data Analysis.
Supervision : Frédéric Chazal

2017 **M1 Internship** (4 months), Simon Fraser University, Vancouver, Canada.
Bioinformatics : clustering (Tuberculosis).
Supervision : Cedric Chauve and Leonid Chindelevitch

2016 **L3 Internship**, ENS Paris-Saclay, Cachan, France.
Numerical Hydrology.
Supervision : Jean-Michel Morel and Marc Lebrun

Education and Degrees

2019–2022 **PhD Thesis**, Université Paris-Saclay and Inria Saclay.
Contributions to the statistical analysis of graph structured data.
Advisors : Frédéric Chazal and Pascal Massart.

2015–2019 École Normale Supérieure Paris-Saclay.
- 2019 **ENS Paris-Saclay Diploma.**
- 2018 **Master's Degree**, *Probability and Statistics,* Université Paris-Saclay.

2012–2015 **Preparatory Classes**, Lycée Faidherbe, Lille.

Publications

Journal Articles

2025 **PASCO (PArallel Structured COarsening) : an overlay to speed up graph clustering algorithms**.
E. Lasalle*, R. Vaudaine*, T. Vayer*, P. Borgnat, R. Gribonval, P. Gonçalves, M. Karsai.
Machine Learning.
Journal | arxiv:2412.13592

2023 **Eve, Adam and the Preferential Attachment Tree**.
A. Contat†, N. Curien†, P. Lacroix†, E. Lasalle†, V. Rivoirard.†
Probability Theory and Related Fields.
Journal | arxiv:2303.04752

2021 **Heat diffusion distance processes : a statistically founded method to analyze graph data sets**.
E. Lasalle.§
J. of Applied and Computational Topology, special issue *Data Science on Graphs*.
Journal | arxiv:2109.13213

Preprints

2025 **Topological Spatial Graph Coarsening**.
A. Calissano†, E. Lasalle†. (Submitted)
arxiv:2512.24327

2025 **Joint reproduction number and spatial connectivity structure estimation via graph sparsity-promoting penalized functional**.
E. Lasalle*, B. Pascal.
Accepted at the *International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2026*.
arxiv:2509.20034

2023 **Compressive Recovery of Sparse Precision Matrices**.
T. Vayer*, E. Lasalle*, R. Gribonval, P. Gonçalves.
arxiv:2311.04673

National Conference Proceedings

2025 **A multilevel approach to accelerate the training of Transformers**.
G. Lauga*, M. Chaumette, E. Desainte-Maréville, E. Lasalle, A. Lebeurrier.
GRETSI, Strasbourg, 2025
arxiv:2504.18590

Notes

2025 **A note on the relations between mixture models, maximum likelihood and entropic optimal transport**.
T. Vayer*, E. Lasalle.
arxiv:2501.12005

*. First author or main co-authors with equal contributions. Remaining authors listed in alphabetical order.
†. Authors listed in alphabetical order, in accordance with common practice in Mathematics.
§. Single author, in accordance with common practice for PhD work in Mathematics.

Talks

International Conferences

Sep. 2025 **European Conference on Machine Learning (ECML) 2025**, Porto, Portugal.
PASCO (PArallel Structured COarsening) : an overlay to speed up graph clustering algorithms.
Peer-reviewed conference.

June 2024 **Popnets Workshop**, Copenhagen, Denmark.
Statistical comparison of graph-structured data and its application to distribution shift detection.
Invited speaker.

Jan. 2023 **Workshop on Random Geometry**, CIRM, Luminy, France.
Finding Adam in the nearest-neighbor tree.
Invited speaker.

National Conferences

May 2024 **SFdS Statistics Days**, Bordeaux.
Compressive recovery of sparse precision matrices.
Acceptance based on abstract.

Nov. 2023 **MIA Workshop : Dimensionality Reduction for Learning and Visualization**, Lyon.
Compressive recovery of sparse precision matrices.
Acceptance based on abstract.

June 2022 **SFdS Statistics Days**, Lyon.
Statistical analysis of graphs via heat diffusion processes.
Acceptance based on abstract.

Oct. 2021 **Young Probabilists and Statisticians Colloquium**, Île d'Oléron.
Statistical analysis of graph-structured data via heat diffusion processes.

Dec. 2021 **Young Mathematicians Forum**, Besançon.
Statistical analysis of graph-structured data via heat diffusion processes.

Invited Seminars and Working Groups

Nov. 2025 **Probability and Statistics Seminar, LAREMA**, Angers.
Reproduction number and spatial connectivity estimation via sparsity-promoting functional.

Apr. 2025 **Paris Statistics Seminar**, Paris.
Compressive recovery of sparse precision matrices.

Apr. 2025 **Applied Mathematics Seminar, LMJL**, Nantes.
Compressive recovery of sparse precision matrices.

March 2025 **Statistics Seminar, IRMAR**, Rennes.
Compressive recovery of sparse precision matrices.

Feb. 2025 **Probability and Statistics Seminar, LAMA**, Marne-la-Vallée.
Compressive recovery of sparse precision matrices.

Dec. 2024 **COMPACT Team Seminar, IRISA**, Rennes.
Compressive recovery of sparse precision matrices.

Nov. 2024 **MALT Team Seminar, IRISA**, Rennes.
Statistical comparison of graph-structured data and its application to distribution shift detection.

Sept. 2024 **SIMS Team Seminar, LS2N**, Nantes.
Compressive recovery of sparse precision matrices.

May 2023 **DATA Department Seminar**, LJK, Grenoble.
Statistical comparison of graph-structured data and its application to distribution shift detection.

Jan. 2023 **EcoNet Project Meeting**, Agro Paris-Saclay Campus.
Statistical comparison of graph-structured data.

Nov. 2022 **Working Group, Probability–Statistics Team, LMO**, Orsay.
Testing SBM vs Erdős–Rényi, with L. Martins-Bianco and Z. Naulet.

June 2022 **Machine Learning and Signal Processing Seminar**, ENS, Lyon.
Heat diffusion distance processes for graphs and their application to distribution shift detection.

June 2022 **Celeste Team Seminar, LMO**, Orsay.
Heat diffusion distance processes for graphs and their application to distribution shift detection.

May 2022 **Working Group, Probability–Statistics Team, LMO**, Orsay.
Density estimation from unweighted k-nearest neighbor graphs. With A. Contat and N. Curien.

May 2022 **Datashape Team Seminar**, Porquerolles.
Detecting distribution shifts using activation graphs from neural networks

March 2022 **Working Group, Probability–Statistics Team, LMO**, Orsay.
Presentation of *Identifying the deviator*, arxiv:2203.03744

Oct. 2021 **Working Group, Probability–Statistics Team, LMO**, Orsay.
Presentation of *Finding Adam in random growing trees*, arxiv:1411.3317

March 2021 **PhD Outreach Seminar**, Université Paris-Saclay.
Gaussian approximations for random functions.

Awards and Funding

International Grants

2024 **Short-Term Exchange Grant** awarded by the **CNRS–Imperial "Abraham de Moivre" International Research Laboratory** for a project with Anna Calissano (*Imperial*, now *UCL*).

Awards

2021 **Winners of the Math–Industry Challenge (AMIES)**. With O. Hacquard and V. Lebovici.
Trajectory reconstruction from noisy position data.
AMIES Challenge

Research Activities

International Research Visits

Nov. 2024 **Research visit to London (1 week)**, collaboration with Anna Calissano (*Imperial*, now *UCL*)
Funding : *Short-Term Exchange Grant* awarded by the *CNRS–Imperial "Abraham de Moivre" International Research Laboratory*.

International Editorial Activities

2025 Review for **IEEE Transactions on Signal and Information Processing over Networks**.
2025 Review for **ALEA** (Latin American Journal of Probability and Mathematical Statistics).
2024 Review for **Bernoulli**.

National Editorial Activities

2025 Reviews for the French-speaking signal processing conference, **GRETSI** 2025.
page of the event
2023 Review for the French-speaking signal processing conference, **GRETSI** 2023.
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Scientific Service and Events

Sep. 2025 - Organization of the **Machine Learning Reading Group** of the SIMS team (LS2N).
Sep. 2023 Organization of the IXXI one-day workshop : **Frugality and Machine Learning**.
Page of the event

Scientific Mediation and Popularization

May 2023 **High School Outreach Talk.** Lycée A. Ribot, Saint-Omer, France.
What does it mean to be a researcher in mathematics?
Presentation covering academic path, research areas, and daily life of a researcher.

Jan. 2021 **Scientific Mediation** with the *Maison d'Initiation et de Sensibilisation aux Sciences* (Université Paris-Saclay). Design and facilitation of science–society debate sessions for high school students.

Teaching

École Centrale Nantes (Adjunct)

2025–2026 **Course Coordinator** (lectures, practical sessions, organization of a data challenge)
Data Science for the Energy Sector, Engineering students (2nd–3rd year), ENRRES track.

2025–2026 **Teaching Assistant** (practical sessions, grading)
Scientific Computing and Optimization, Engineering students (2nd–3rd year), DATASIM track.

2025–2026 **Teaching Assistant** (practical sessions, grading)
Mathematical Tools for Signal and Image Processing, Master 2 CORO DASSIP.

ENS de Lyon (Adjunct)

2024–2025 **Teaching Assistant** (practical sessions, project supervision)
Complex Networks, Master 2 Complex Systems.

IUT d'Orsay – Computer Science (Teaching Fellowship)

2019–2022 **Teaching Assistant** (tutorials, practical sessions, grading)
Modeling (linear algebra, diagonalization, Python labs), 2nd-year students.

2019–2022 **Teaching Assistant** (tutorials, grading)
Probability and Statistics (standard discrete and continuous distributions, approximation theorems, CLT, estimators, statistical tests), 2nd-year students.

2019–2022 **Teaching Assistant** (tutorials, grading)
Discrete Mathematics (logic, basics of linear algebra), 1st-year students.

Tutoring

2017–2018 **Tutoring** three students from the Villebon–Charpak Institute, Orsay.
Design of refresher courses and exercises for academic support at the L1–L2 level.

Skills

Programming and Computing

- Python, Github
- networkx, GUDHI
- LaTeX, ipe

Languages

French	●●●●●
English	●●●●○
German	●○○○○